EINDHOVEN UNIVERSITY OF TECHNOLOGY

Robot Foosball Table

Research question: What limits the interception performance of a foosball robot built on an IR touch frame and differential BLDC rod drives: sensing latency, prediction error or actuator response? And how much does compensating the dominant limit improve the defense rate as ball speed increases?

Description

Table football is fast and unpredictable, so a robot that plays it shows the limits of sensing, prediction and actuation. The hardware is a foosball table with an IR touch frame for ball sensing and a differential BLDC drive that moves and rotates a rod. This project focuses on what limits the robot's ability to intercept the ball: sensing latency, prediction error or actuator response.

A single rod drive will be evaluated first, followed by measuring the end to end latency of the ball perception. Next, to build a Kalman filter based ball state predictor with and without a wall reflection model. Each predictor will be tested with closed loop on the rod to measure block rate against ball speed.

A ROS 2 framework will also be integrated into the system to help with modular control and visualization. If the first rod drive works well, three more will be built, so one controller can run four rods with a simple strategy - the rod nearest to the ball moves to it and shoots. The long term goal is a robot that can play against human opponents.

Learning Goals

The student can

- Review earlier work on automated foosball tables and ball tracking.
- Test the IR touch frame (rate, latency, jitter, accuracy, drop outs) using an independent timing reference.
- Mount the differential rod drive on the table, use an EtherCAT motor driver and control the actuation.
- Implement a Kalman filter ball state predictor.
- Design motion control for the differential drive that positions the rod at the target setpoint and decouples translation from rotation.
- Integrate the software in a ROS 2 framework for modular control and visualization.
- Run closed loop experiments with repeatable shots, measure defense rates against ball speed with and without prediction.
- Run the controller on four rods and have a play session against human players (stretch)